A source-bounded transit interface for reducing cognitive load in local bus use

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Live bus trackers expose route, vehicle, stop, and prediction data, but a rider may still need to translate those data into a personal action. We built a location-first WRTA interface that binds saved Worcester destinations to nearby stops, low-walk direct or one-transfer bus options, route geometry, current location, and walking estimates on one stable map. The contribution is not a new transit authority data source. It is a source-bounded interaction layer that separates official stop-time predictions from derived destination-arrival estimates, suppresses impractical long-walk options, reduces map interpretation work, and makes uncertainty visible for a rider who wants fewer choices and less context switching.

1 Introduction

Public transit information systems often contain the necessary facts without presenting the specific fact a rider needs in the moment. The official Worcester Regional Transit Authority (WRTA) tracker provides the live vehicle and route surface for WRTA service (*1*, *2*). For a rider deciding whether to leave, wait, board, or keep watching the map, however, the tracker still leaves a translation step: identify the relevant stop, connect that stop to the destination, decide which route direction matters, infer whether the moving vehicle is the one to catch, and convert an arrival prediction into a personal action.

The AO Labs bus tracker was built as a narrower interface for that translation. The public version at 'bus.aolabs.io' keeps a focused live-bus surface for familiar WRTA routes and six fixed Worcester-area destinations: 96 William Street, Alden Hall, Union Station, Chipotle on Park Avenue, Cold Stone on Main Street, and Blackstone Theaters. It uses WRTA live and stop-time surfaces, Ride Guide route, stop, and schedule geometry, OpenStreetMap tiles, browser geolocation, Leaflet rendering, and OSRM walking estimates (*3*–*9*). The app is therefore not authoritative beyond the sources it reads. It is a personal layer that narrows those sources into one map and a short set of low-walk trip options.

The design motivation is neurodivergent fit. The intended rider has self-reported autism and ADHD and wants the interface to remove repeated interpretation work rather than ask for more manual planning. This paper therefore treats cognitive load as an engineering requirement: keep the map visible, keep the screen stable, name the stop, show the route, show the live bus, distinguish stop time from destination time, and avoid alarm-like interaction patterns. General web accessibility standards also emphasize predictable, operable, and understandable interfaces, but this paper makes a narrower claim: a personal transit layer can be better for a specific rider by reducing avoidable interpretation (*10*).

2 Results

2.1 A stable, location-first map changes the task

The WRTA tracker is route-first: routes and vehicles are the primary objects. The AO Labs interface is location-first: a selected place reframes the map around the rider's current location, the selected destination, the boarding stop chosen from that current location, any required transfer, the exit stop near the destination, route geometry, and live buses when browser GPS is available (Fig. 1). This changes the task from inspecting the transit network to answering a personal question: which stop, route sequence, and bus time matter for this trip? When direct options require a long walk, the transfer search can surface a two-bus route instead of filling the panel with high-walk single-route alternatives.

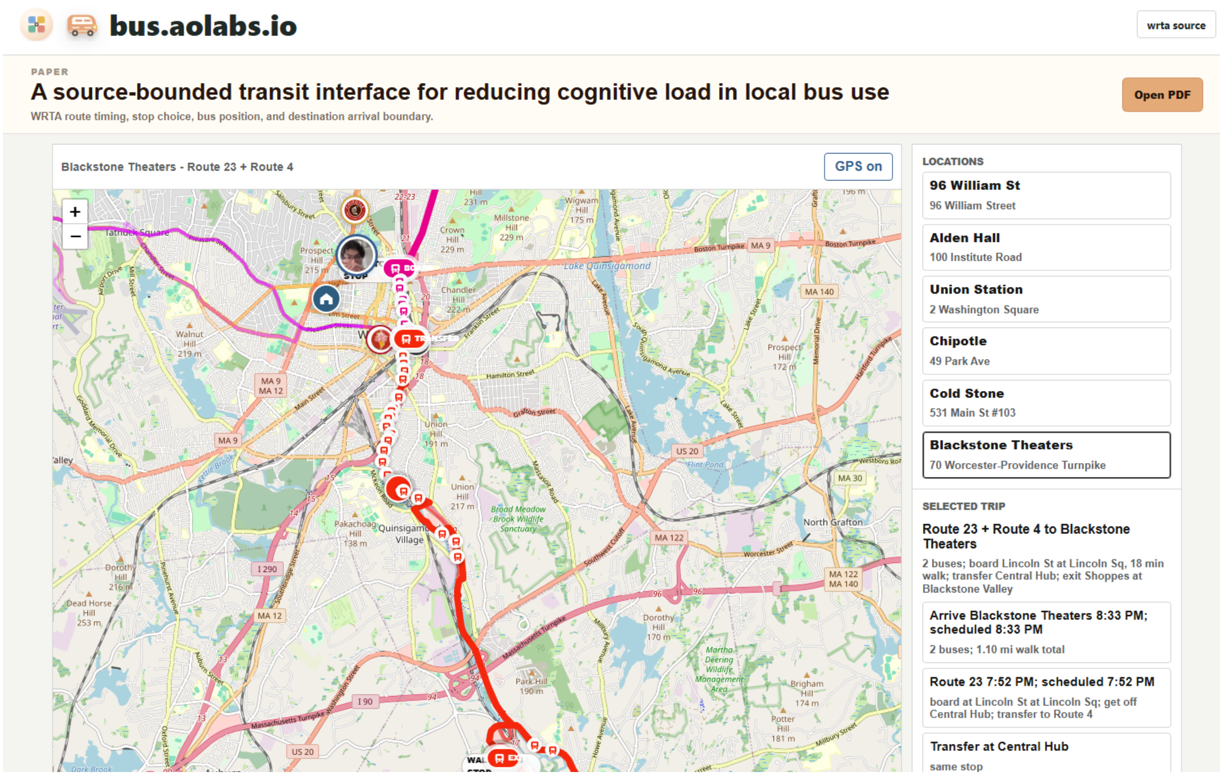

Figure 1. Stable live interface for selected WRTA destinations. The public app keeps saved destinations, route geometry, bus-stop symbols, live-bus state, and current-location state on one map while the side panel keeps selected-trip state in the same view. In this capture, simulated browser geolocation near WPI with Blackstone Theaters selected surfaces a Route 23 to Route 4 transfer through Central Hub. The app suppresses high-walk Route 4-only alternatives and separates the bus-at-stop times from the derived destination-arrival estimate.

Table 1. Comparison of the official source surface and the personal source-bounded layer.

Task	WRTA tracker	AO Labs bus tracker
Primary object	Route and system map	Saved place or selected destination
Stop selection	Interpreted by rider	GPS-based boarding stop, transfer stop, exit stop, or labeled fallback
Location context	Not personalized by default	Current location shown when browser GPS is available
Timing row	Stop prediction surface	Bus-at-stop time, transfer sequence, and derived destination estimate
Map behavior	General live operations view	Selected-place framing with route, transfer, and stop context
Authority	Official WRTA surface	WRTA-derived, route-bounded, not authoritative beyond sources

2.2 The app separates bus-at-stop time from destination-arrival time

The most important timing correction is semantic. A bus can arrive at a stop, and the rider can arrive at the selected destination later. If the interface collapses those events into one timestamp, the rider has to guess what the time refers to. The current app displays both values as separate claims. 'Bus predicted' and 'scheduled' are derived from WRTA stop-time output for the selected boarding stop when GPS is available. The destination-arrival value is computed by adding scheduled in-vehicle travel and OSRM exit-stop-to-place walking duration, with live delay propagated when a WRTA prediction is available.

This separation prevents the app from overstating certainty. The first value is a transit-source value for a stop. The second is a derived estimate for a place. The second depends on walking route availability, walking speed, business entrance geometry, traffic signals, and whether the rider starts walking immediately after leaving the bus. The UI labels it as approximate, and the paper treats it as a derived interface estimate rather than an official WRTA prediction.

2.3 The design removes repeated interpretation work

The app is better than a general tracker for this personal use case because it removes specific repeated burdens. The rider does not need to count vague map distances, identify a stop from an unlabeled route line, infer whether a location pin is connected to a route, decide whether a long walk is being substituted for a missing transfer, or wait for a static bus marker to jump after the next feed update. The app renders route paths, bus-stop symbols, transfer context, live buses, place icons, and current location together, and it interpolates live bus markers between source updates so the vehicle appears to move continuously while still depending on WRTA data.

The interface also avoids disruptive alerts. It does not use vibration, audio, popups, emergency warnings, or alarm-like notifications. This is not a cosmetic choice. For a rider who is already tracking a bus, walking, and managing uncertainty, a calm persistent screen can be more usable than a system that interrupts attention.

2.4 Bounded superiority over the source tracker

The superiority claim is intentionally bounded. The WRTA tracker remains the official and broader source. The AO Labs app is superior only for a narrower problem: turning a small set of personal destinations into route, stop, bus, and timing context without asking the rider to interpret the full transit system. Its advantage comes from scope reduction and stable spatial framing, not from more authoritative predictions.

3 Discussion

This work shows that a transit interface can improve utility by reducing interpretation rather than adding features. For a rider with ADHD and autism, the difficult step is often not obtaining data; it is holding multiple spatial and temporal facts in working memory while the map changes. A personal tracker can reduce that cost by keeping the relevant facts on the same screen and by labeling what each time actually means.

The app also demonstrates a useful boundary for AI- or automation-adjacent personal tools: the system should not invent certainty. The correct design is not to promise exact destination arrival. It is to show the stop prediction, show the walking basis, compute a clear derived estimate, and keep the source boundary visible. This makes the app calmer and more trustworthy because the rider can see which parts are official data and which parts are personal context.

Several limitations remain. The public app currently focuses on six saved destinations, a focused live-bus surface, and direct or one-transfer low-walk trip options rather than full-system itinerary optimization. The current default hides auto-selected options over 1.5 miles of total walking and searches same-day service only; this is appropriate for the intended rider but not a universal transit-planning objective. Browser geolocation may be blocked or unavailable depending on device, browser, operating system, and permission state. WRTA stop predictions can be absent, stale, or changed by operations. OSRM walking routes may not match the exact path a rider chooses. No controlled user study has yet measured missed buses, time saved, stress, or reduction in interpretation errors. Until those observations exist, the paper presents the app as a source-bounded product and design record, not a population-level accessibility result.

4 Materials and Methods

4.1 System

The app is a Node.js web service deployed at 'https://bus.aolabs.io/'. The browser client renders a Leaflet map with OpenStreetMap tiles. Saved destinations are fixed latitude-longitude records in the application source. The server exposes JSON endpoints for route geometry, live vehicles, trip planning, selected-location arrivals, walking routes, destination geocoding, and health state.

4.2 Transit sources

Live vehicle state is read from the WRTA SWIV service. Stop arrivals are read from WRTA stop-time output for the selected stop. Route stops, route shapes, and schedule details are read through Ride Guide endpoints for the WRTA dataset. The app does not create official transit predictions; it displays or derives values from those upstream sources.

4.3 Boarding-stop selection

For a selected saved destination, the primary path uses browser GPS as the trip origin. The server evaluates direct trips and one-transfer trips from WRTA route candidates, searches feasible boarding, transfer, and exit stop combinations, and ranks options with walking distance as the primary burden. Auto-selected options are filtered to same-day service and a 1.5-mile total-walking ceiling before they are shown. The UI displays the boarding stop from the current location, the transfer stop when needed, the exit stop near the selected place, route context, and upcoming stop arrivals. If GPS is unavailable, the app uses a labeled location-nearest fallback instead of implying that the fallback is the rider's boarding stop.

4.4 Destination-arrival estimate

For each selected trip, the server obtains upcoming stop times for each selected boarding stop and requests OSRM walking routes from the current location to the first boarding stop, between transfer stops when needed, and from the exit stop to the selected destination. Destination arrival for a two-bus trip is calculated as

$$T_{\mathrm{destination}} = T_{\mathrm{board1}} + D_{\mathrm{ride1}} + D_{\mathrm{transfer}} + W_{\mathrm{transfer}} + D_{\mathrm{ride2}} + D_{\mathrm{exit\ walk}},$$

where T_{board1} is the WRTA-derived first boarding-stop time, D_{ride1} and D_{ride2} are scheduled in-vehicle travel times, D_{transfer} is the scheduled wait after the transfer walk, W_{transfer} is the transfer walking duration, and $D_{\mathrm{exit\ walk}}$ is the OSRM walking duration from the exit stop to the selected place. Direct trips are the same expression with the transfer terms removed. If OSRM fails, the server falls back to a straight-line walking estimate at the configured walking speed of 1.34 m s^{-1}. The UI labels destination arrival as a source-bounded derived estimate rather than an official WRTA prediction.

4.5 Visual verification

The screenshot in Figure 1 was generated from the local app at desktop width using simulated browser geolocation near WPI with Blackstone Theaters selected. This verifies that the public interaction model has the expected ingredients for a same-day transfer state: current-location marker, selected destination, live-bus layer, selected-trip panel, route sequence, transfer stop, and suppression of long-walk single-route options.

5 Acknowledgements

The app was shaped by repeated personal corrections from Alan Pham around map framing, stop labeling, icon recognition, timing ambiguity, GPS visibility, and the need for a stable low-interruption screen.

6 Funding

No external funding supported this work.

7 Author contributions

Alan N. Pham specified the use case, evaluated the interface, and authored the design requirements. AO Labs implemented the public app, generated the manuscript artifact, and performed verification.

8 Competing interests

The author declares no competing interests.

9 Data availability

The public interface depends on WRTA live vehicle and stop-time outputs, Ride Guide route and stop geometry, OpenStreetMap tiles, browser geolocation, and OSRM walking estimates. The live surface is available at ‘https://bus.aolabs.io/’.

10 Code availability

The code is maintained in the AO Labs bus app repository. Public repository access is not currently advertised on the app page.

11 Additional information

This manuscript is a product and design record for a personal transit interface. It does not present a controlled user study, clinical result, official WRTA prediction system, or guarantee of arrival-time accuracy.

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